

$$h(u) = 4, \quad h(v) = 5$$

SUPREMUM AND INFIMUM

Q.1 (30)

Let S be a partially ordered set, and let A be a subset of S . An element M in S is called an upper bound of A if M succeeds every element of A , that is, for every $x \in A$, we have

$$x \leq M$$

If an upper bound of A precedes every other upper bound of A , then it is called the supremum of A and it is denoted by

$$\sup(A)$$

We also write $\sup(a_1, \dots, a_n)$ instead of $\sup(A)$ when A consists of the elements $a_1, \dots, a_n$. We emphasize that there can be at most one $\sup(A)$; however, $\sup(A)$ may not exist.

MLB

Analogously, an element m in S is called a lower bound of a subset A if m precedes every element of A , that is, for every $y \in A$, we have

$$m \leq y$$

MLB

If a lower bound of A succeeds every other lower bound of A , then it is called the infimum of A and it is denoted by

$$\inf(A)$$

We also write $\inf(a_1, \dots, a_n)$ instead of $\inf(A)$ when A consists of the elements $a_1, \dots, a_n$. Similarly, there can be at most one $\inf(A)$ although $\inf(A)$ may not exist.

MLB

Some texts use the term least upper bound instead of supremum and then write $\text{lub}(A)$ instead of $\text{sup}(A)$, and use the term greatest lower bound instead of infimum and then write $\text{glb}(A)$ instead of $\inf(A)$.

MLB

If A has an upper bound we say A is bounded above, and if A has a lower bound we say A is bounded below. In particular, A is bounded if A has an upper and lower bound.

EXAMPLE 7.7

Existence of $\sup(a, b)$ and $\inf(a, b)$

m16

Let S be an ordered set and let $a, b \in S$. If S is linearly ordered, then $\sup(a, b)$ and $\inf(a, b)$ clearly exist. Specifically, if $a \leq b$, then $\sup(a, b) = b$ and $\inf(a, b) = a$. On the other hand, if S is an arbitrary ordered set, then $\sup(a, b)$ and $\inf(a, b)$ need not exist. However, there are important examples of nonlinearly ordered sets where $\sup(a, b)$ and $\inf(a, b)$ do exist for every a, b in the set.

(31) Lecture

7.8 ISOMORPHIC (SIMILAR) ORDERED SETS

Suppose X and Y are partially ordered sets. A one-to-one (injective) function $f: X \rightarrow Y$ is called a *similarity mapping* from X into Y if f preserves the order relation, that is, if the following condition holds for any pair $a, b \in X$:

$$a \leq b \text{ in } X \text{ if and only if } f(a) \leq f(b) \text{ in } Y$$

The above condition is equivalent to the following two conditions:

- $$\left[\begin{array}{l} (1) \text{ If } a \leq b \text{ then } f(a) \leq f(b). \\ (2) \text{ If } a \parallel b \text{ (noncomparable), then } f(a) \parallel f(b). \end{array} \right]$$

Accordingly, if the underlying sets X and Y are both linearly ordered, then only (1) is needed for f to be a similarity mapping.

Two ordered sets X and Y are said to be *order-isomorphic* or *isomorphic* or *similar*, written

$$\underline{X \simeq Y}$$

if there exists a one-to-one correspondence (bijective mapping) $f: X \rightarrow Y$ which preserves the order relations, i.e., which is a similarity mapping. Such a function f is then called an *order-isomorphism* from X onto Y or an *order-isomorphism* between X and Y .

7.10 LATTICES

Let L be a nonempty set closed under two binary operations called *meet* and *join*, denoted respectively by $\wedge$ and $\vee$. Then L is called a *lattice* if the following axioms hold where a, b, c are any elements in L :

[L₁] **Commutative law:**

$$(1a) \ a \wedge b = b \wedge a$$

$$(1b) \ a \vee b = b \vee a$$

[L₂] **Associative law:**

$$(2a) \ (a \wedge b) \wedge c = a \wedge (b \wedge c)$$

$$(2b) \ (a \vee b) \vee c = a \vee (b \vee c)$$

[L₃] **Absorption law:**

$$(3a) \ a \wedge (a \vee b) = a$$

$$(3b) \ a \vee (a \wedge b) = a$$

We will sometimes denote the lattice by $(L, \wedge, \vee)$ when we want to show which operations are involved.

Duality and the Idempotent Law

The *dual* of any statement in a lattice $(L, \wedge, \vee)$ is defined to be the statement that is obtained by interchanging $\wedge$ and $\vee$. For example, the dual of

$$\underline{a \wedge (b \vee a) = a \vee a} \quad \text{is} \quad \underline{a \vee (b \wedge a) = a \wedge a} \quad \text{m.c.g.}$$

Notice that the dual of each axiom of a lattice is also an axiom. Accordingly, the principle of duality holds; that is:

Theorem 7.6 (Principle of Duality): The dual of any theorem in a lattice is also a theorem.

This follows from the fact that the dual theorem can be proven by using the dual of each step of the original theorem.

An important property of lattices follows directly from the absorption laws.

Theorem 7.7 (Idempotent Law): (i) $a \wedge a = a$, (ii) $a \vee a = a$.

The proof of (i) requires only two lines:

$$\begin{aligned} a \wedge a &= a \wedge (a \vee (a \wedge b)) && \text{(using (3b))} \\ &= a && \text{(using (3a))} \end{aligned}$$

WELL-ORDERED SETS

(32) Lecture

Not every ordered set, even if it is linearly ordered, need have a first element. On the other hand, one of the fundamental properties of the set

$$\mathbb{P} = \{1, 2, 3, \dots\}$$

of counting numbers (positive integers) is that $\mathbb{P}$ and every subset of $\mathbb{P}$ has a first element. Such an ordered set is said to be well-ordered. Namely:

Definition 8.1: Let A be an ordered set. Then A is said to be *well-ordered* if every subset of A contains a first element.

Note that any well-ordered set A is linearly ordered. For if $a, b \in A$, then the subset $\{a, b\}$ of A has a first element which, therefore, must precede the other; hence any two elements of A are comparable.

The following theorem follows directly from the above definition.

Theorem 8.1: Let A be a well-ordered set. Then:

- (i) Every subset of A is well-ordered.
- (ii) If B is similar to A , then B is well-ordered.

Theorem 8.3: ... are similar to each other.

8.3 TRANSFINITE INDUCTION

The reader is familiar with the principle of mathematical induction. Namely:

Principle of Mathematical Induction: Let S be a subset of the set P of counting numbers with the following two properties:

- (1) $1 \in S$.
- (2) If $n \in S$, then $n + 1 \in S$.

Then S is the set of all counting numbers, that is, $S = P$.

The above principle is one of Peano's axioms for the counting numbers P . The principle can be shown to be a consequence of the fact that P is well-ordered. In fact, there is a somewhat similar statement which is true for any well-ordered set (proved in Problem 8.1).

Theorem 8.4 (Principle of Transfinite Induction): Let S be a subset of a well-ordered set A with the following two properties:

- (1) $a_0 \in S$.
- (2) If $s(a) \subseteq S$, then $a \in S$.

Then S is the entire set A , that is, $S = A$.

Here a_0 is the first element of A and $s(a)$, called the *initial segment* of a , is defined to be the set of all elements in A which strictly precedes a .

Initial segments will be discussed below, and Chapter 9 will discuss transfinite induction in much more detail.

9.2 CARTESIAN PRODUCTS AND CHOICE FUNCTIONS

The following theorem applies.

Definition 9.1: Let $\{A_i : i \in I\}$ be a nonempty family of nonempty sets. Then the cartesian product $\{A_i : i \in I\}$, denoted by

$$\prod \{A_i : i \in I\} \text{ or } \prod_i A_i$$

is the set of all choice functions defined on $\{A_i : i \in I\}$.

Recall that a function $f: \{A_i : i \in I\} \rightarrow X$, where each A_i is a subset of X , is called a choice function if $f(A_i) = a_i$ belongs to A_i , for every $i \in I$. In other words, f "chooses" a point $a_i \in A_i$ for each set A_i .

EXAMPLE 9.1 Let $\{A_1, A_2, \dots, A_n\}$ be a finite family of sets. Recall (Chapter 2) that the cartesian product of the sets,

$$A_1 \times A_2 \times \dots \times A_n = \prod_{i=1}^n A_i$$

is defined to be the set of n -tuples

$$(a_1, a_2, \dots, a_n)$$

where $a_i \in A_i$ for $i = 1, 2, \dots, n$. On the other hand, each choice function f defined on $\{A_1, A_2, \dots, A_n\}$ corresponds to an n -tuple $(a_1, a_2, \dots, a_n)$ where $a_i = f(A_i)$.

AXIOM OF CHOICE

The axiom of choice lies at the foundations of mathematics and, in particular, the theory of sets. This "innocent looking" axiom, which follows, has as a consequence some of the most powerful and important results in mathematics.

Axiom of Choice: The cartesian product of a nonempty family of nonempty sets is nonempty.

Using Definition 9.1, the axiom of choice can be stated as follows:

Axiom of Choice: There exists a choice function for any nonempty family of nonempty sets.

The axiom of choice is equivalent to the following postulate:

Zermelo's Postulate: Let $\{A_i : i \in I\}$ be any nonempty family of disjoint nonempty sets. Then there exists a subset B of the union $\bigcup_i A_i$ such that the intersection of B and each set A_i consists of exactly one element.

Observe that in Zermelo's postulate the sets are disjoint whereas in the axiom of choice they may not be disjoint.

WELL-ORDERING THEOREM, ZORN'S LEMMA

The following theorem is attributed to Zermelo, who proved the theorem directly from the axiom of choice.

Well-Ordering Theorem: Every set can be well-ordered.

Zorn's lemma, which follows, is one of the most important tools in mathematics; it establishes the existence of certain types of elements although no constructive process is given to find these elements.

Zorn's Lemma: Let X be a nonempty partially ordered set in which every chain (linearly ordered subset) has an upper bound in X . Then X contains at least one maximal element.

We formally state and prove (Problem 9.4) the following basic result of set theory:

Theorem 9.1: The following are equivalent:

- (i) Axiom of choice;
- (ii) Well-ordering theorem;
- (iii) Zorn's lemma.